CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 2: Requirement Analysis

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 2: Requirement Analysis**
Student Name:	Arnaov Garg	**Email:**	ag5088@srmist.edu.in
Institution:	SRM Institute of Science and Technology (SRMIST)	**Track / Cohort:**	Snowflake Data Engineering (Snowflake Data Engineering_EL_2027_Cohort 1)
GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Functional Requirements (FR)

Requirement ID	Specification & Scope
FR-1: Automated Data Ingestion	System must stage and ingest 5 operational CSV files (Customers, Restaurants, Menu Items, Orders, Order Items) into Snowflake RAW tables via internal stages without manual schema manipulation.
FR-2: Data Cleansing & Transformation	Staging ETL must enforce type safety (dates, timestamps, decimals), trim whitespace, standardize casing, sanitize phone/email records, and handle null prices.
FR-3: Operational Outcome Tagging	System must derive binary flags (is_delivered, is_delivered_late, is_cancelled) and compute realized_revenue (zero for cancelled orders) to guarantee accurate financial reporting.
FR-4: Dimensional Star Schema Modeling	Curated layer must establish 2 fact tables (FACT_ORDERS, FACT_ORDER_ITEMS) and 5 dimension tables linked via MD5 surrogate keys.
FR-5: Feature Engineering & Segmentation	Pipeline must engineer analytical features: RFM loyalty tiers (VIP, Frequent, Occasional, Single), restaurant rating tiers, menu price tiers, and meal periods (Breakfast, Lunch, Snacks, Dinner, Late Night).
FR-6: Persona Analytical Data Marts	System must deliver 3 pre-aggregated views: V_PRIYA_CUSTOMER_ENGAGEMENT, V_DANIEL_RESTAURANT_PARTNERSHIPS, and V_GRACE_EXECUTIVE_OVERVIEW.
FR-7: Interactive Streamlit UI	Web portal must provide interactive charts (line, bar, scatter, pie) with global dynamic filtering by Date Range, Operational City, Cuisine, and Order Status.

2. Non-Functional Requirements (NFR)

Requirement ID	Specification & Scope
NFR-1: Sub-Second Query Latency	Analytical queries executed by the dashboard must return in under 200 ms on an X-Small virtual warehouse.
NFR-2: Micro-Partition Pruning	Snowflake queries filtered by temporal ranges (e.g., single month) must prune over 85% of micro-partitions.
NFR-3: Storage & Cost Efficiency	Virtual warehouse ORDERPULSE_WH must auto-suspend after 60 seconds of inactivity to minimize credit consumption.
NFR-4: Data Completeness & Integrity	Zero records dropped during ETL; 100% referential integrity enforced through surrogate primary/foreign key relationships.
NFR-5: Security & Layer Isolation	Physical separation of database schemas (RAW, STAGING, CURATED) to maintain zero modification access to landing tables.

3. Software & Technical Environment Specifications

Component	Technology / Specification	Role in Pipeline
Cloud Data Warehouse	Snowflake Cloud Data Platform (Standard/Enterprise)	Storage, compute warehouse, ELT transformations & query pruning
ETL & Modeling Language	Snowflake SQL (DDL, DML, Window Functions)	Three-tier schema transformations and analytical view materialization
Application Framework	Streamlit 1.66 / Python 3.12	Multi-tab interactive dashboard and stakeholder exploration UI
Data Analytics Libraries	Pandas 3.0, Plotly Express 7.1, NumPy	In-memory metric aggregations and chart rendering
Version Control	Git & GitHub Repository	Code repository, SQL versioning, documentation, and deployment